PURDUE UNIVERSITY | Elmore Family School of Electrical and Computer Engineering

ECE 49022 – Electrical Engineering Senior Design Projects
Fall 2026 – Product Development Plan – Update 2

MASHD – Multifunction Appliance for Smart Home Display

Written by: Alexander Beal, Javier Aldana, Joe McLoughlin, Issa Haddad

1.0 Project Information

1.2 Team Number: 20

2.2 Project Title: MASHD - Multifunction Appliance for Smart Home Display

3.2 Instructor Team Advisor: Shivam Duhan

4.2 GTA Team Mentor: Surya Vishal Bodireddy

5.2 Team Members and Roles:

Name	Major CmpE or EE	Expected Graduation	Team Role
Alexander Beal	**EE**	**December 2026**	**Team Lead**
Javier Aldana	**EE**	**December 2026**	**Hardware Lead**
Joe McLoughlin	**CmpE**	**May 2027**	**Software Lead**
Issa Haddad	**CmpE**	**May 2027**	**Systems Lead**

2.0 Project Abstract

The MASHD device is a multifunction smart home specifically tailored to college kids and reducing noise complaints while providing maximum parallel use cases. Functionality includes a decibel meter, clock display, temperature display, Bluetooth device and Wi-Fi connectivity, an onboard speaker, ambient light sensor, infrared person detection sensor, and TV control. This solution will use an app to configure the functionality of the MASHD device as well as notify the user when any set limit is reached.

3.0 Proposal and Concept Development

3.1 Problem Statement

College kids aren't considered great neighbors, they are noisy, go to sleep late and throw parties even on Wednesdays. Because of the undeveloped frontal lobes inside of every twenty-one-year-olds brain, they are very susceptible to the modern tragedy called noise complaints. As more Purdue students are forced to move off campus due to on campus housing limitations, this becomes an ever-growing problem for West Lafyette's residents, equally those who want to go to sleep at a reasonable time and those who want to live it up during their college years. According to one Exponent Article from 2008, [1] "The majority of the city's complaints are issued to Purdue students ... between five and 20 violators per month". This issue has probably gotten bigger since then and is not constrained to Purdue; campuses across the country face these issues as admission rates rise and campus housing struggles to keep up. Is the answer to this another single use gadget to use once and throw away? Here lies the second issue, people own too many things. From personal experience, we at Team 20 do not necessarily want another single function gadget in our homes.

3.2 Proposed Solution

Our proposed solution is a multifunction decibel meter with the main objective of reducing college apartment noise complaints while providing maximum functionality as a college apartment wall fixture. Functionality includes a decibel meter, clock display, temperature display, Bluetooth device and Wi-Fi connectivity, an onboard speaker, ambient light sensor, infrared person detection sensor, and a barrel jack connector power. This solution will use an app to configure the functionality of most of the above functions as well as notify the user when the set decibel level is exceeded.

Breakdown of functionality:

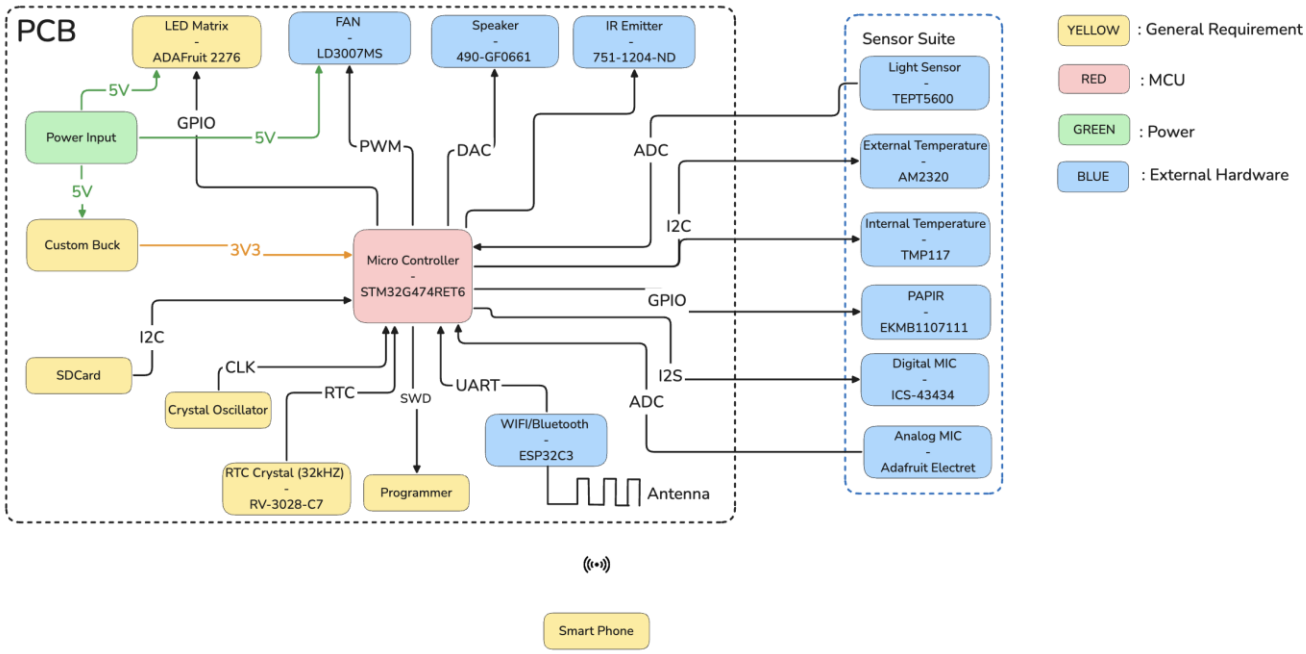
- Decibel meter: Measure the noise level of the room (Main functionality)
- Dot matrix display: Display decibel, clock, temperature, and any other displayable data
- Clock: Provide the user a time reference since acceptable noise levels vary with time of day
- Speaker: Notify the user of an exceeded noise level. Customizable sound / tone
- Bluetooth / Wi-Fi connectivity: Provide the user a wireless and convent way to customize device parameters such as max decibel level, speaker tones, and display look. Also provides an additional way to notify users of an exceeded noise level
- Additional functionality
    - Ambient light sensor: Dim display brightness according to room ambient light brightness
    - Infrared person detection: Detect presence of people in the room, turning the device off if room is empty, and activating display if people are detected
    - Temperature measurement: Provide room temperature on display
    - Powered from barrel jack: Commonly used power plug

3.3 Customers/Stakeholders

The main appeal of this product is towards the college / young adult market. For those who consistently receive noise complaints, and a broad estimate using rough information available places that over 1 million college students will encounter law enforcement due to a noise

complaint filed by their neighbors [1]. This is the key feature that spawned the ideation of this product, as having a way to determine when the party / room is too loud that there is risk of a noise complaint. This could be used in college dorms and apartments to establish a baseline level through failures to set a threshold for when police may get called for noise complaints, which is helpful to know as a host of a party if you need to tell everyone to quiet down. There are additional features that are nice to have like calendar and other 'smart' features, such as weather, customization, and clock capabilities, as well as interior monitoring, as the appeal is to be a complete device as a collection of sensors, the only one you will need.

Diagram 1: Block Diagram



3.4 Project-Level Functional Description

The device primarily shall measure sound level and compare against predetermined thresholds such to alert the users if the sound level exceeds the predetermined value.

The user shall be alerted through a display

- **Decibel meter**: Measure the noise level of the room (Main functionality)
- **Dot matrix display**: Display decibel, clock, temperature, and any other displayable data
- **Clock**: Provide the user a time reference since acceptable noise levels vary with time of day
- **Speaker**: Notify the user of an exceeded noise level. Customizable sound / tone
- **Bluetooth / Wi-Fi connectivity**: Provide the user a wireless and convent way to customize device parameters such as max decibel level, speaker tones, and display look. Also provides an additional way to notify users of an exceeded noise level

Additional functionality

- **Ambient light sensor**: Dim display brightness according to room ambient light brightness
- **Infrared person detection**: Detect presence of people in the room, turning the device off if room is empty, and activating display if people are detected

- **Temperature / Humidity measurement**: Provide room temperature on display
- **USB-C PD**: 5V power for device

4.0 Design Requirements

4.1 Project-Level Engineering Design Requirements

1) The product shall be clearly visible from across the room to someone with 20-20 vision or corrective lenses.
2) The product display shall be settable by the user from their mobile device.
3) The product should have a low power mode setting in the case of no person being present.
4) The product shall be wirelessly connected to a mobile phone.
5) The product shall monitor human activity, noise, light, temperature, and humidity.
6) The product shall be powered over a cable coming from a wall outlet.
7) The product shall supply the necessary voltages at the necessary currents for every component.
8) The product shall be wall mountable.
9) The product shall have appropriate mounts and holes for all the internal electrical components.
10) The product shall maintain temperatures below maximum operating conditions of 60ºC.
11) The product should pull weather data, calendar information, live flight data, and other internet-enabled features from the internet.

S1) The product could flash firmware updates over Wi-Fi to all compute modules.
S2) The product could set images beyond strings to the display.
S3) There product could have an additional admin interface present that is password protected

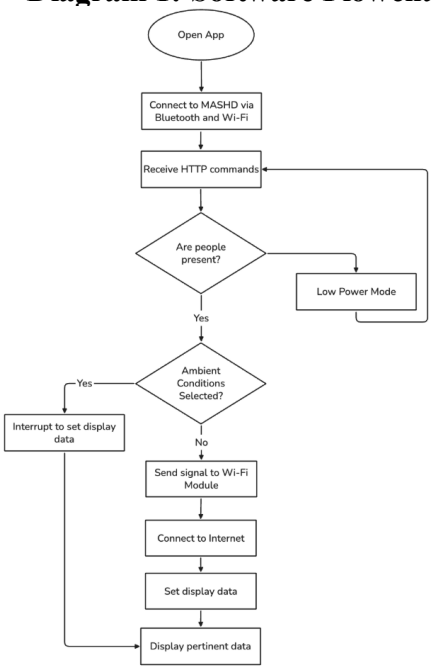
4.2 Subsystem-Level Engineering Design Requirements

4.2.1 User Display
4.2.2 Processing
4.2.3 Wireless
4.2.4 Sensor Suite
4.2.5 Power
4.2.6 Mechanical

5.0 System Design

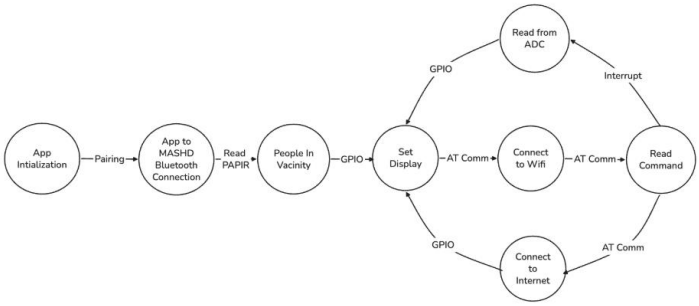
5.1 Software Overview

Diagram 1: Software Flowchart



The main software protocol is to connect from your phone / web browser to the device, and then commands will be sent to the communications chip (ESP 32), which will pass on the commands to the STM 32. If we do not see people present, the display will enter a low power state, turning down / off the display in the Low Power Mode. If there are people present, we will poll the internet for updated weather data and the latest commands, and set the display state with interrupts if the user chooses that screen.

Diagram 2: Software State Machine



Our overall state machine pairs the app to the device via Bluetooth or Wifi, which then goes and polls the PAPIR sensor, to set the display state, poll internet information, or update sensor values.

5.2 Mechanical Overview

The MASHD CAD design features a rectangular, open-back enclosure that supports three LED matrices while keeping the display area unobstructed. The frame includes mounting holes for the matrices, an internal PCB mounting plate, and dedicated openings for the sensors, USB-C connection, fan, and speaker. Ventilation slots allow airflow through the enclosure. The prototype will be 3D-printed in sections, while the final housing will use plywood with finger-jointed corners.

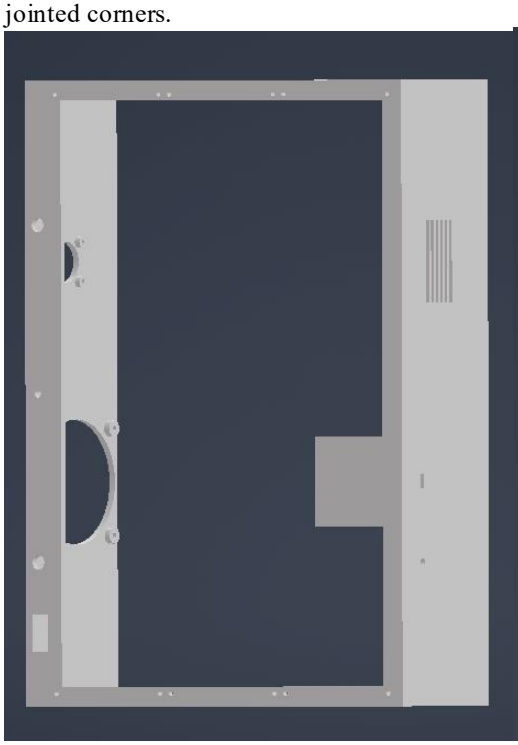


Figure 1: front side of MASHD (290X380x85mm)

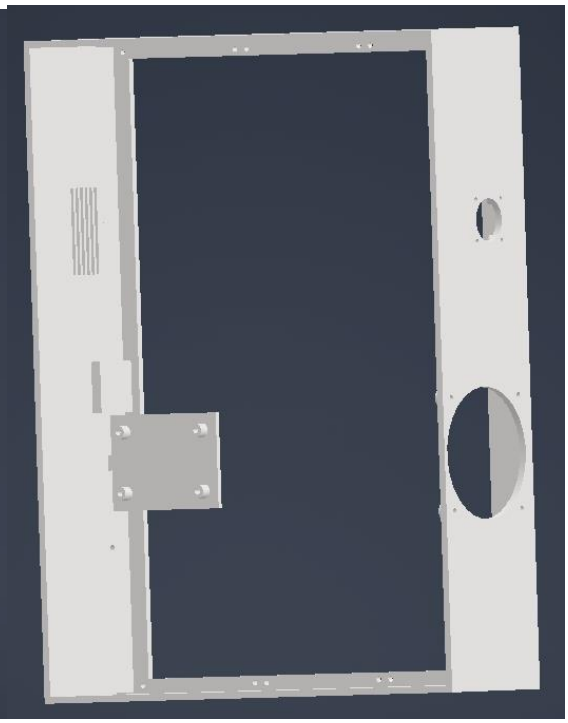
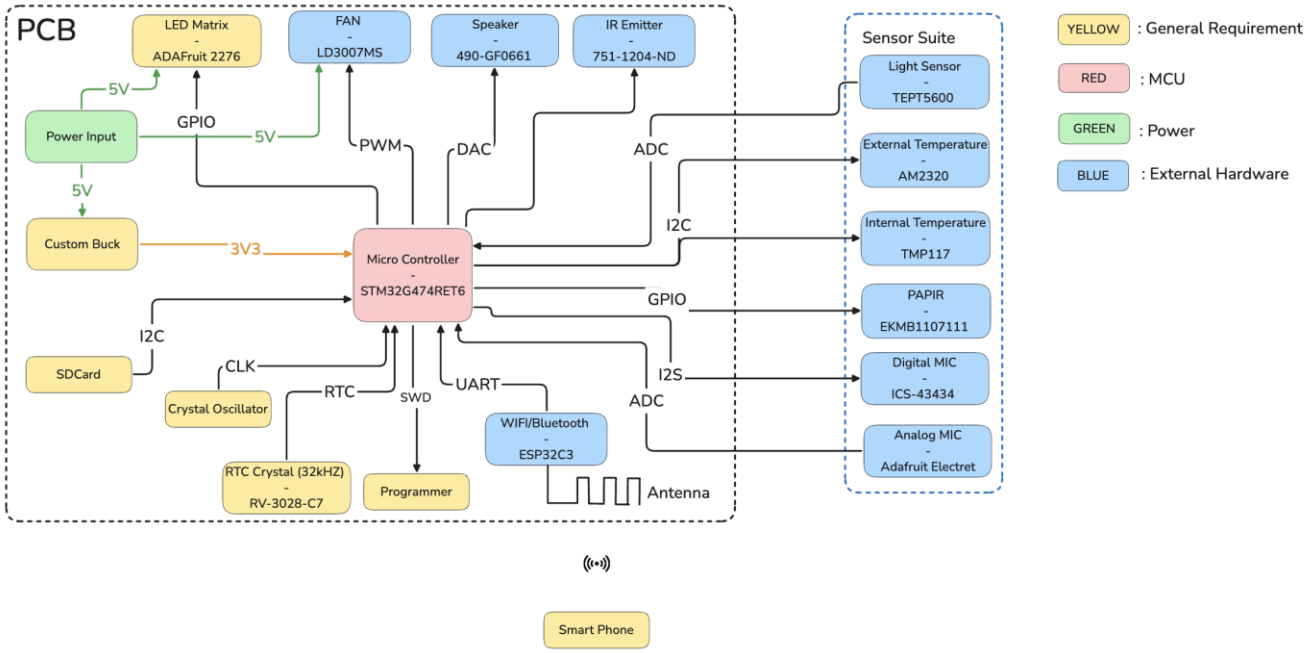


Figure 2: back side of MASHD (290x390x85mm)

Material/component	Quantity	Estimated weight	Estimated cost
Plywood enclosure frame with finger joints	4 frame pieces	580g	N/A
Wood glue for frame joints	x1	N/A	N/A
M3 LED matrix and PCB mounting screws	16 screws (12 matrix; 4PCB)	14.1g	$5.48
M2.5 × 16 mm fan mounting screws and nuts	4 screws + 4 nuts	3.72 g	$6.64
M2.5 Speaker mounting screws and nuts	4 screws + 4 nuts	2.6g	$4.00
Fixed flat wall-mounting brackets	2 brackets	150g	$9.99
Bracket-to-frame wood screws	4 screws	16.8g	$9.08
Bracket-to-wall screws	4 screws	28g	$2.48
Drywall anchors, if required	4 anchors	10g	$3.50

5.3 Electrical Overview



Overall system block diagram: All of the peripherals are connected to the STM32, which runs its own code to poll the sensors and update screen values, however it receives commands from the ESP 32 via UART (and eventually more data via SPI), for what to display and to reflect the changes made by the user on the web server or the app. Through these interfaces, the ESP 32 parses the input from the web server or iOS app, and sends the appropriate command via UART, so that the communications are processed completely separately from the sensor / screen values to allow for independent systems that will not impact one another if they are having issues.

5.4 Software Formalization

6.0 Testing and Verification

1.2 Base-Product Design Requirement Testing/Verification Plans

2.2 Stretch Design Requirement Testing/Verification Plans

7.0 Budget and Resources

Project Materials Budget Table:

8.0 Development Plan and Timeline

8.1 Individual Project Design Reports

The following table shows which team member is responsible for writing each individual project design report.

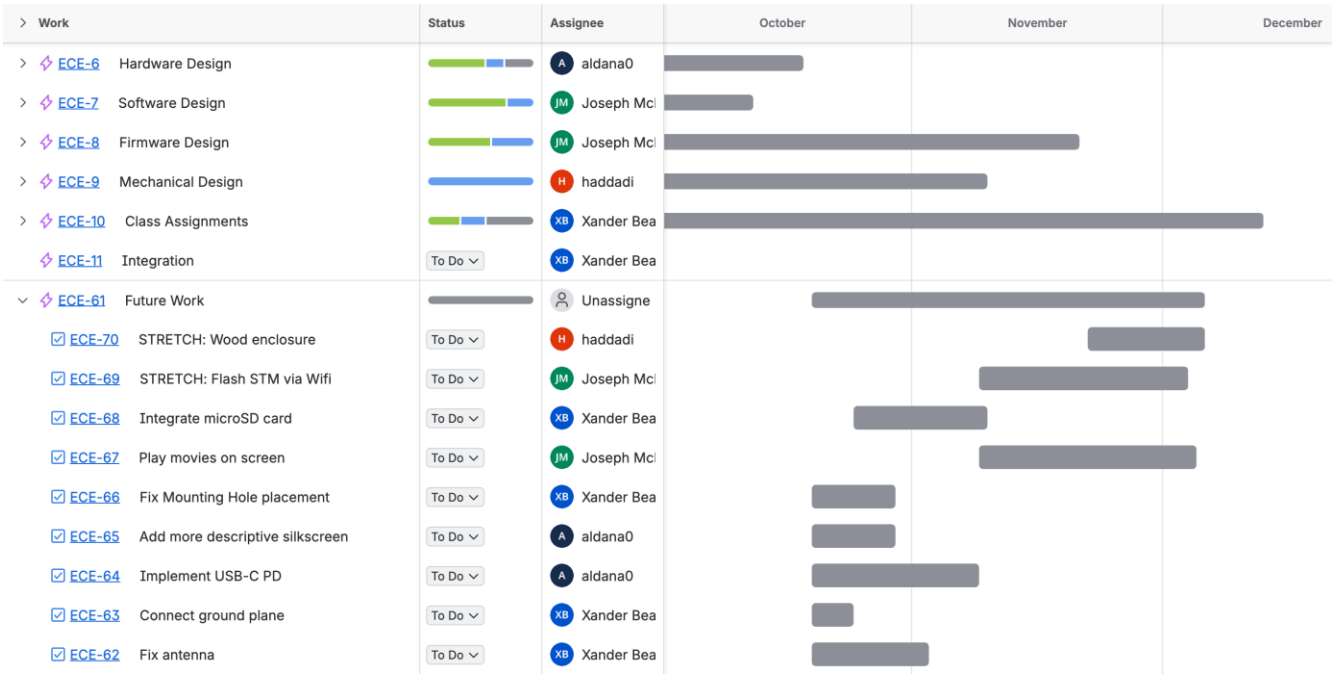
Design Report	To be written by:
Software Overview	Javier Aldana
Electrical Overview	Alexander Beal
Mechanical Overview	Issa Haddad
Software Formalization	Joe Mcloughlin

8.2 Individual Project Analysis Reports

The following table shows which team member is responsible for writing each individual project analysis report.

Analysis Report	To be written by:
Legal and Regulatory Analysis	Javier Aldana
Reliability and Safety Analysis	Alexander Beal
Ethical and Environmental Analysis	Issa Haddad
User Manual	Joe McLoughlin

8.3 Project Timeline



9.0 Sources Cited

[1]"L1: Smoothing Input," *Physical Computing*, 2025. https://makeabilitylab.github.io/physcomp/advancedio/smoothing-input.html (accessed Sept. 22, 2026).

[2] NTCLE400E3472H Vishay Beyschlag/draloric/BC ..., https://www.digikey.com/en/products/detail/vishay-beyschlag-draloric-bc-components/NTCLE400E3472H/2230794 (accessed Sep. 23, 2026).

[3] Adafruit Industries, "64x32 RGB LED Matrix - 6mm pitch," *adafruit*, 2020. https://www.adafruit.com/product/2276 (accessed Sept. 18, 2026).

[4] "At Command set□," ESP, https://docs.espressif.com/projects/esp-at/en/latest/esp32/AT_Command_Set/index.html (accessed Sep. 23, 2026).

[5] "Analysis of universal asynchronous receiver-transmitter(uart) | IEEE conference publication | IEEE xplore," Analysis of Universal Asynchronous Receiver-Transmitter(UART), https://ieeexplore.ieee.org/document/10532820/ (accessed Sep. 24, 2026).

[6] P. Burgess, "Adafruit protomatter RGB matrix library," Adafruit Learning System, https://learn.adafruit.com/adafruit-protomatter-rgb-matrix-library/overview (accessed Sep. 25, 2026).

[7] "Ti," A Basic Guide to I2C, https://www.ti.com/lit/an/sbaa565/sbaa565.pdf?ts=1681380825945 (accessed Sep. 25, 20

[1] A. Svensson, "Campus Safety Statistics," *Worldmetrics.org*, Feb. 12, 2026. https://worldmetrics.org/campus-safety-statistics/ (accessed Sept. 10, 2026).

Appendix A: PCB Bill of Materials Tables

Main PCB Bill of Materials